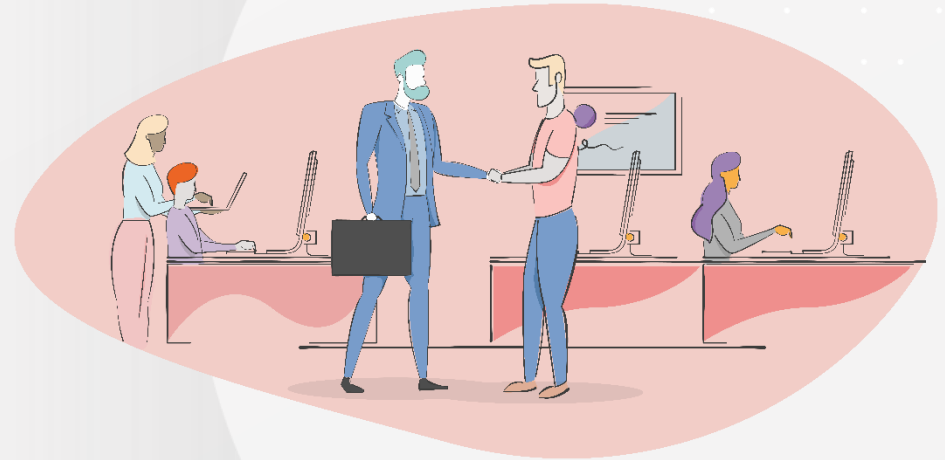


Introduction to Data Quality



L5 Data Engineer Higher Apprenticeship
Module 1 / 12 (“Data Fundamentals”)
Topic 2 / 4

The value of data quality

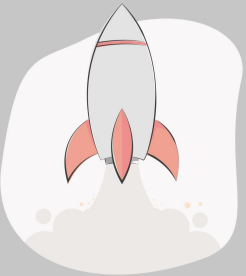
Case study example: Novartis

The problem - Novartis faced data management and integration challenges across various sources

The solution:

- Adopted FAIR Data Principles and implemented data governance framework
- Implemented Novartis Data Catalog, a metadata management solution like OpenMetadata

The benefits:



- Improved data findability, accessibility, interoperability, and reusability
- Enhanced data quality and consistency
- Increased collaboration and data sharing
- Efficient data discovery and access
- Better regulatory compliance



Novartis Pharmaceutical Corporation

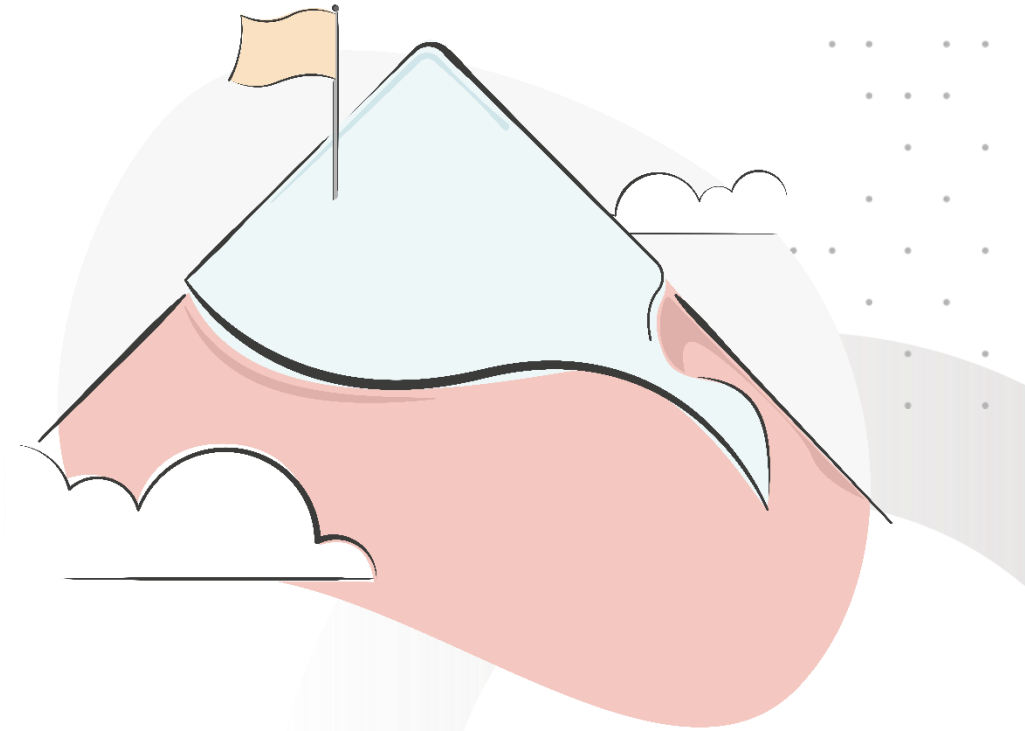
Image source: Reuters

Learning outcomes

This webinar is designed to support the following learning outcomes:

- Define data quality and its significance
- Demonstrate an understanding of the importance of data standards in data engineering
- Apply data quality management strategies proficiently to improve the accuracy and reliability of data
- Analyse sustainable data practices and governance principles to advocate for long-term data quality and integrity

Building Careers
Through Education



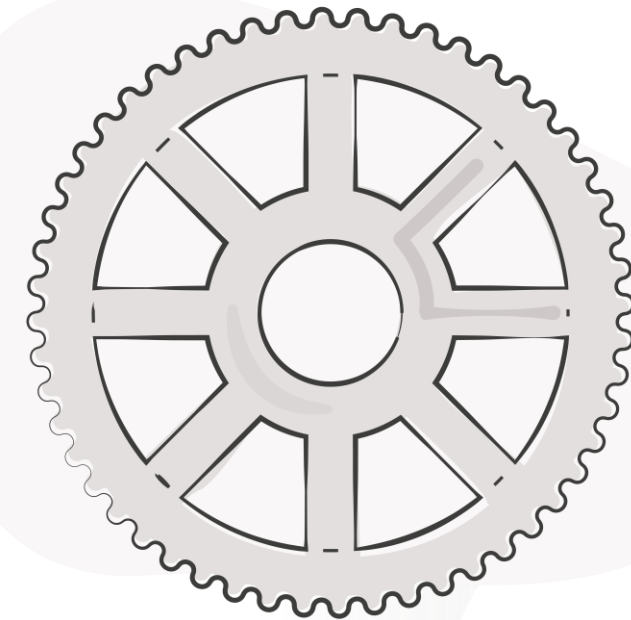
Webinar agenda

This webinar will cover the following:

- Data and metadata standards
- Navigating data quality issues
- Responsible data handling – best practices
- Common strategies for ensuring data quality
- Why focusing on quality helps sustainability
- **Practical: Data Quality Testing**

Webinar length: 3 hours

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Data quality challenges

Discussing your previous experience...

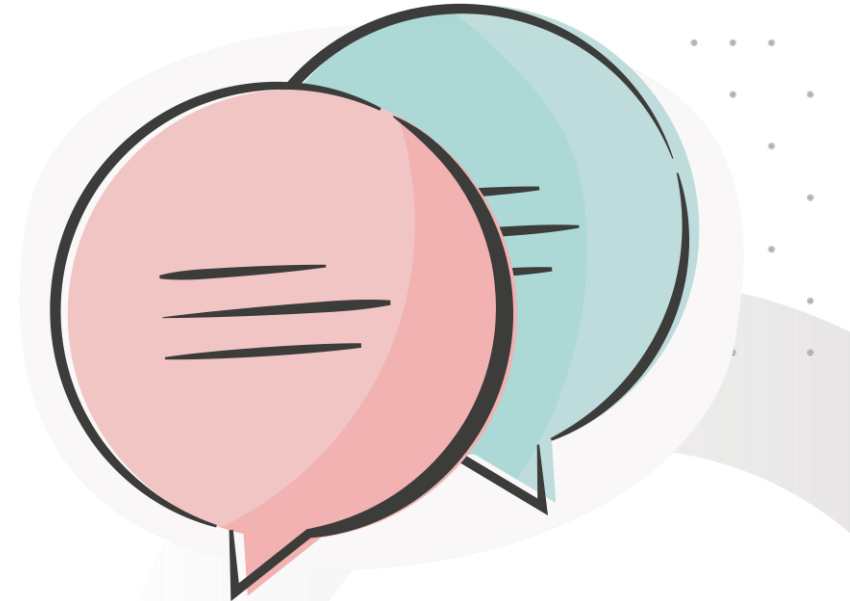
Let's discuss your prior knowledge and experiences with data quality challenges.

Share any instances where you or your organisation faced issues related to data accuracy, completeness, consistency, or timeliness.

Consider the following questions:

- How did these data quality challenges impact your work or decision-making processes?
- What measures, if any, were taken to address these challenges?

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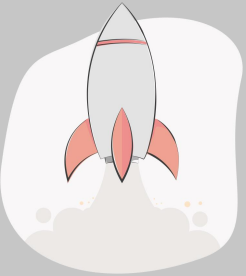


Group discussion

Module Case Study

A reminder

- Credit Bank Corporation aimed to gain insights into employee performance, satisfaction, and HR processes
- The HR team planned to build a comprehensive analytics dashboard to track key metrics
- Data engineers must integrate diverse data and follow best practices
- The team must determine data types, build a suitable data ecosystem, visualise results, and articulate business value



This 'Data Fundamentals' module is designed to provide you with the knowledge and skills to overcome these challenges!



An analytics dashboard

Data and metadata standards



Section Introduction

Defining data standards

Data standards are guidelines for data management practices.

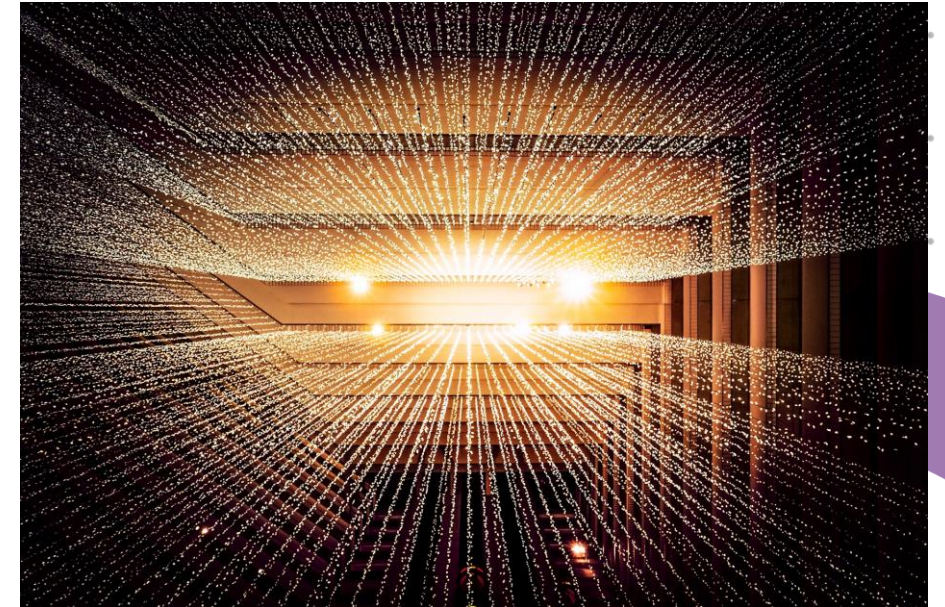
They enhance data reliability and originate from various sources:

- Regulatory bodies
- Standards organisations
- Professional associations and consortia
- Governments
- Corporations
- Academia
- Community initiatives



Standards evolve constantly and vary across industries.

Many data standards are open standards.



Knowledge Check Poll

Let's see what you can remember from the e-learning!

Which of the following is NOT one of the four main data quality metrics discussed in the e-learning?

- A) Accuracy
- B) Integrity
- C) Consistency
- D) Relevance

Feedback: D – Relevance is not one of the four main data quality metrics discussed in the e-learning.

The four main metrics covered were accuracy, integrity, consistency, and timeliness (also known as recency or availability).

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**Submit your responses to
the chat!**



Data quality metrics

An introduction

Data quality encompasses various dimensions (or metrics).

These dimensions or metrics are broken down below:

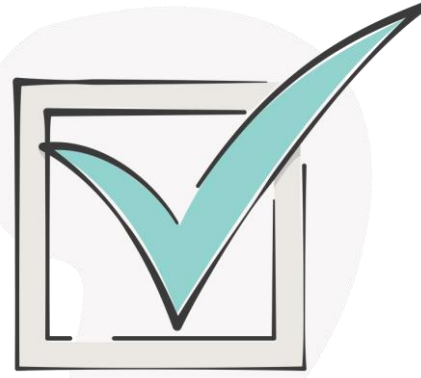
Data quality metric	Alternative names	Answers the questions
Accuracy	Correctness, validity	How well does our data reflect the real world? Are false items likely?
Integrity	Completeness	How well does our data cover the real world? Are missing items likely?
Consistency	Uniformity	How well does our data conform to a single standard? Are discrepancies likely?
Timeliness	Recency, availability	How up-to-date is our data? Are stale items likely?
Reliability	N/A	Is the data consistently accurate over time? Can the data be depended upon to make critical decisions?



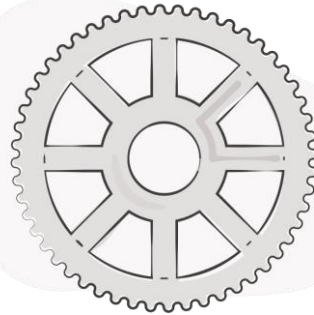
Introducing open standards



Definition: guidelines for technology that anyone can use and contribute to



What they offer: Compatibility, flexibility, innovation and cost-effectiveness



Their importance: Ensure tools and systems are compatible and adaptable for seamless integration and evolution



Examples: HL7 in healthcare, HTTP, OpenAPI



The FAIR data standard

An introduction

FAIR data principles include:

- Findability
- Accessibility
- Interoperability
- Reusability

Defined in a March 2016 paper in Scientific Data by a consortium of scientists and organisations.



Findable



Accessible



Interoperable

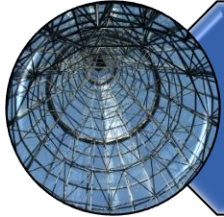


Reusable

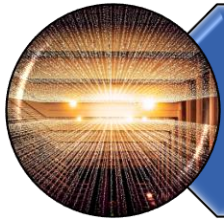
Components of the FAIR standards

Applying the FAIR data standard

What it means in practice...



Provides a framework for data management practices



Improves data quality, accessibility, integration and reusability



Steps need to be taken to make data **findable**, **accessible**, **interoperable**, and **reusable**



Findable



Accessible

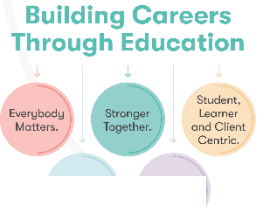


Interoperable



Reusable

Components of the FAIR standard



Applying the FAIR data standard

Group discussion

Credit Bank Corporation's HR department encountered data issues due to inconsistent data practices across the enterprise.

This resulted in inaccuracies in key HR metrics.

Applying FAIR data principles can resolve this challenge.

Consider the following questions:

- How can the bank ensure that HR data is "Findable" by authorised users?
- How can the bank achieve "Interoperability" between HR data and other datasets or systems?

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Through Education



Case study: Credit Bank Corporation



Group
discussion



The Dublin Core Metadata Initiative

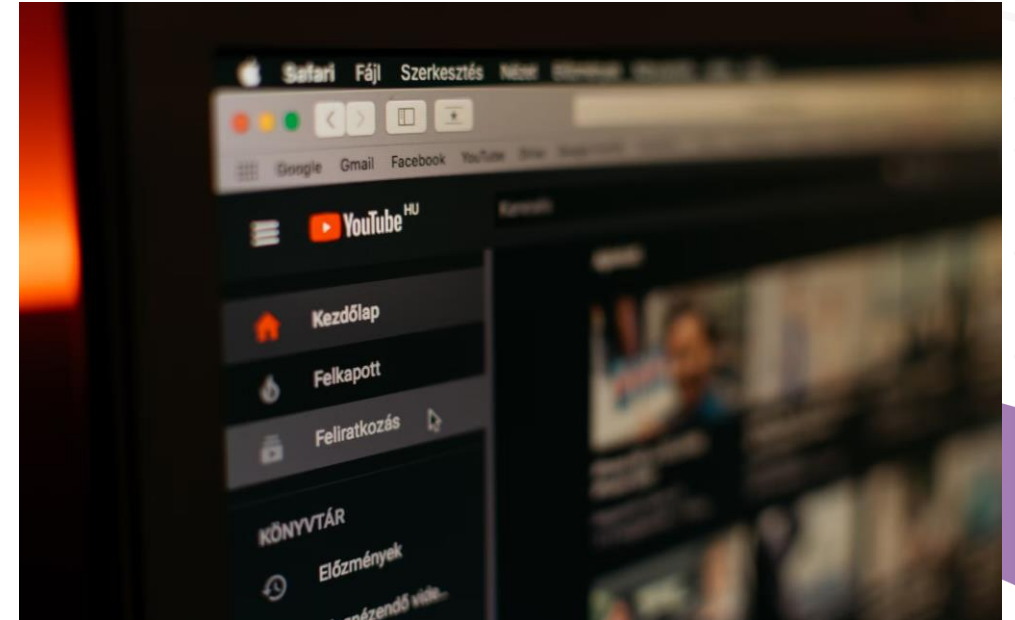
Introducing DCMI

The Dublin Core Metadata Initiative (DCMI) offers a set of metadata standards used to describe resources in various domains.

Dublin Core™ Metadata Initiative



Its core elements provide a basic framework for describing digital resources such as documents, images, videos, and web pages.



The Dublin Core Metadata Initiative

Use cases and implementation

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Through Education



Digital libraries



Archives and museums



Educational resources



Digital publishing



Popular on GOV.UK

→ [HMRC account: sign in or set up](#)

→ [Universal Credit account: sign in](#)

→ [Self Assessment tax return: sign in](#)

Government portals



Unstandardised data

The challenges

In today's abundance of data, organisations face a myriad of challenges when dealing with unstandardised data.

These include the following:



Data silos



Inconsistencies



Integration issues



Unstandardised data

Overcoming the challenges

Standards play a pivotal role in overcoming issues of data siloing, inconsistencies and integration.



Facilitating Data
Sharing



Promoting
Collaboration



Enhancing
Interoperability

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Through Education



Knowledge Check Poll

Which of the following initiatives focuses on developing standards for describing resources for the purpose of finding them?

- A) FAIR data principles
- B) Data Quality Testing Methodologies
- C) Dublin Core Metadata Initiative
- D) Unique Identifiers Management Framework

Feedback: C – The Dublin Core Metadata Initiative (DCMI) is a global organisation that develops standards for describing resources to improve their discoverability.

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the chat!**



Navigating data quality issues



Section Introduction

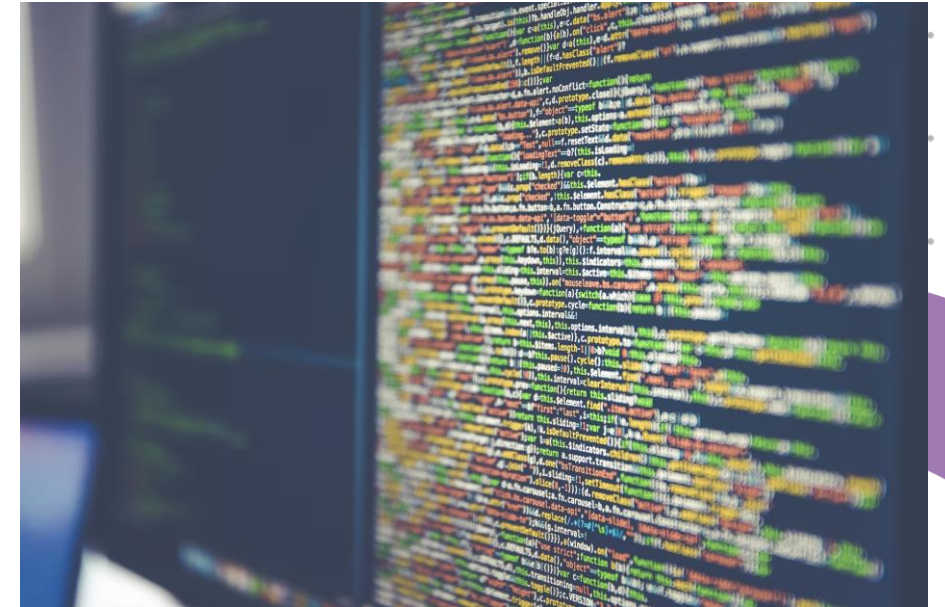
Data formats and quality issues

We will now explore the common issues with the quality of widely-used data formats and how to overcome them, including:

- XML
- CSV
- JSON



Addressing data quality issues with these formats is crucial for enabling seamless data exchange and integration.

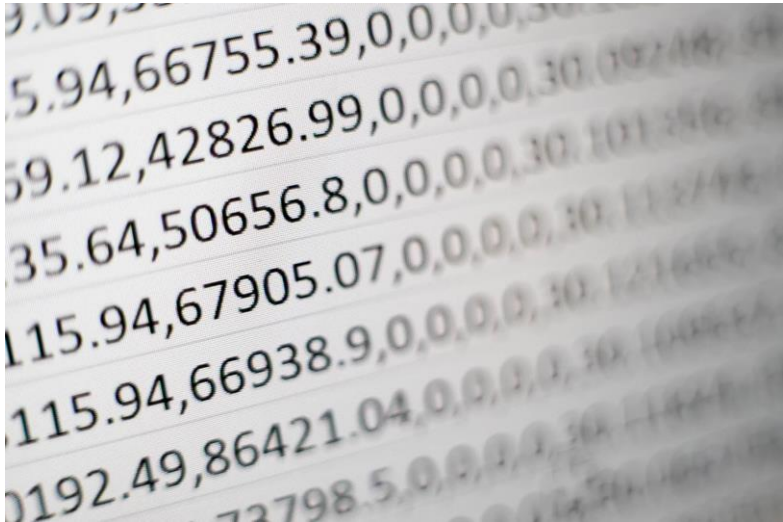


Challenges and quality issues

CSV

While CSV files offer simplicity and portability, they also present several quality challenges. Here is some guidance to help you:

Issue	Description	How can it be mitigated?
Inconsistent formats	Inconsistencies in delimiters, encapsulators, or line terminators	Standardise the use of delimiters, encapsulators, and line terminators, Implement robust parsing algorithms
Data type mismatch	The absence of explicit data types in CSV files	Provide metadata or schema information, perform data validation and type casting during parsing
Header misalignment	Missing or misaligned headers	Ensure consistent headers and alignment, implement error handling mechanisms



The CSV data format

Challenges and quality issues

JSON

While JSON offers several advantages in terms of its lightweight and human-readable format, it also presents a set of challenges.

Here is some guidance to help you:

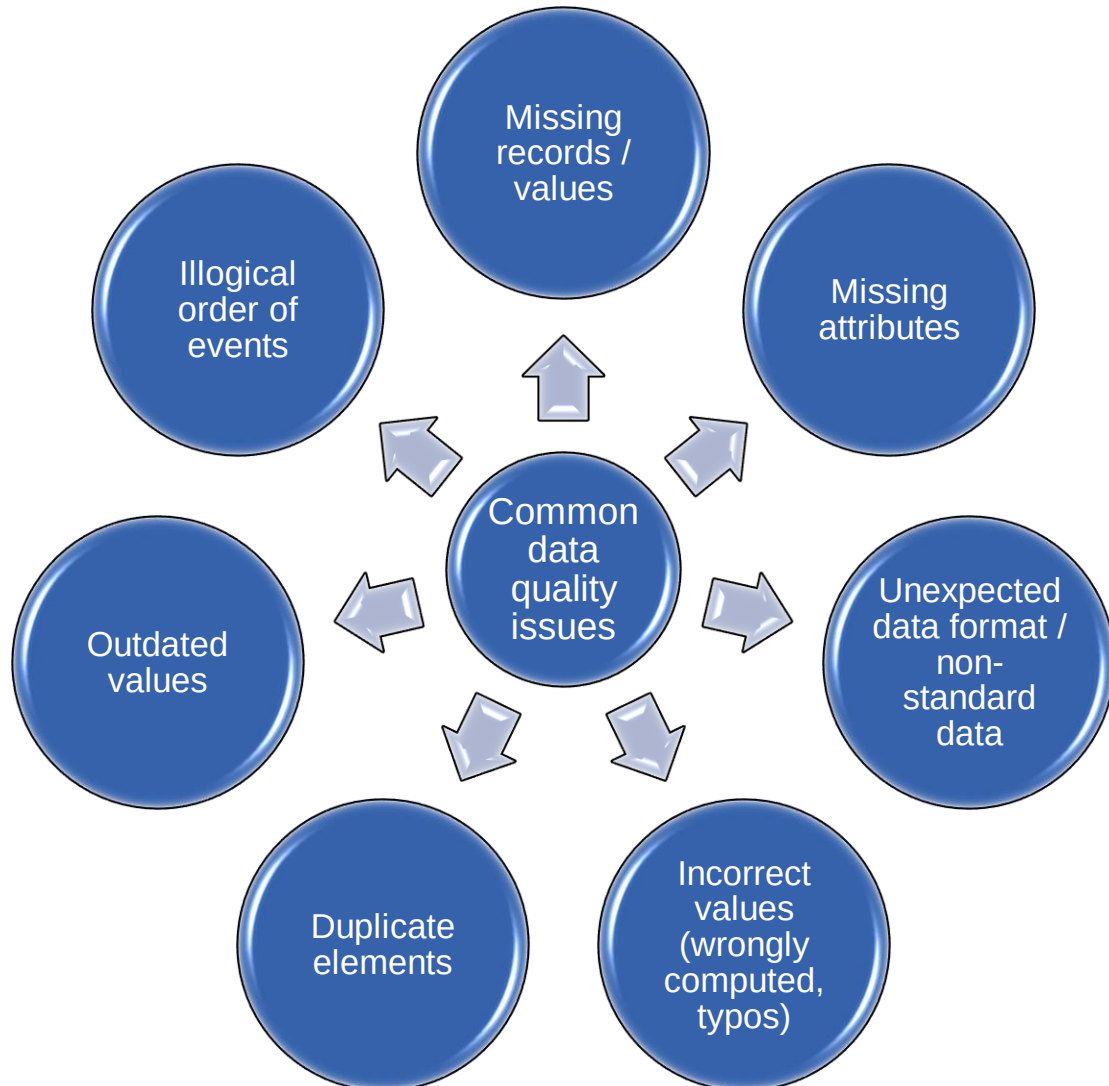
Issue	Description	How can it be mitigated?
Nesting complexity	Nested structures can complicate parsing and increase processing time	Design JSON to balance complexity and efficiency
Key-value pair integrity	Inconsistent key naming or missing/extra keys can lead to errors	Establish naming conventions and validation rules
Data volume	Large files can present challenges for memory usage, processing time, and network bandwidth	Data streaming, pagination, compression, and distributed processing



The JSON data format

Other common data quality issues

Further data quality issues you may encounter include the following:



	id	first_name	last_name	email
▶	1	Carine	Schmitt	carine.schmitt@verizon.net
	4	Janine	Labrune	janine.labrune@aol.com
	6	Janine	Labrune	janine.labrune@aol.com
	2	Jean	King	jean.king@me.com
	12	Jean	King	jean.king@me.com
	5	Jonas	Bergulfsen	jonas.bergulfsen@mac.com
	10	Julie	Murphy	julie.murphy@yahoo.com
	11	Kwai	Lee	kwai.lee@google.com
	3	Peter	Ferguson	peter.ferguson@google.com
	9	Roland	Keitel	roland.keitel@yahoo.com
	14	Roland	Keitel	roland.keitel@yahoo.com
	7	Susan	Nelson	susan.nelson@comcast.net
	13	Susan	Nelson	susan.nelson@comcast.net
	8	Zbyszek	Piestrzeniewicz	zbyszek.piestrzeniewicz@att.net

Duplicate elements example

Applying your learning

Group discussion

The team at Credit Bank Corporation will have to import data from multiple systems to build their HR dashboard, including:

- Enterprise Management Cloud (XML)
- Payroll software (export as CSV)
- 360 Feedback Tool for employee evaluations (exports as CSV)

Consider the following questions:

- What potential XML quality issues may arise when importing from the Enterprise Management Cloud? How can they be mitigated?
- What potential CSV issues may be found when importing payroll and feedback data? How can they be mitigated?

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Case study: Credit Bank Corporation



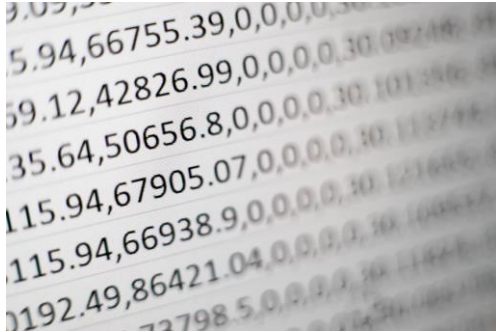
Group
discussion



Importing HR data

Practical top tips!

Follow these tips to ensure data quality and seamless integration when importing HR data in different formats:



The CSV data format

- Watch for data type misinterpretations, define columns explicitly
- Check date/time formats, convert to standard corporate timezone
- Support Unicode for non-Latin text fields like names



The XML data format

- Validate structure/syntax to prevent parsing errors
- Standardize encoding, escape special characters
- Adhere to defined schemas for conformity



The JSON data format

- Design nested structures for complexity/efficiency balance
- Implement strict naming/validation for key-value pairs
- For large files, use streaming, pagination, compression



Common strategies for ensuring data quality



Section Introduction

Data lineage and unique identifiers

Let's now take a look at data lineage and unique identifiers.

These are pivotal for transparent and reliable data management, ensuring integrity and compliance.



Data lineage: the lifecycle of data, encompassing its origin, usage, and movement over time



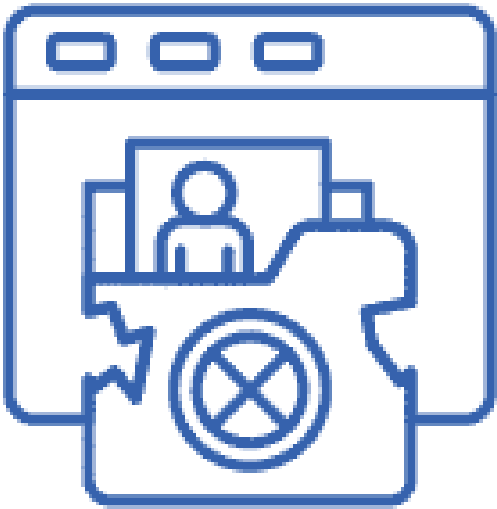
Unique Identifiers: UIDs ensure each data item is distinct, enabling effective data management and retrieval

UIDs vs UUIDs

A comparison

So, what's the difference between Unique Identifiers (UIDs) and Universally Unique Identifiers (UUIDs)?

Here's how to remember them:



UIDs: Unique within their individual system

Vs



UUIDs: UUIDs are globally unique across different systems

Data lineage and unique identifiers

Application in the context of our case study

Credit Bank Corporation planned to build a comprehensive analytics dashboard to track key HR metrics

- In HR dashboards, UUIDs can differentiate employees, track their performance reviews, and manage payroll data securely and efficiently
- UUIDs vs. UIDs: While both serve as unique identifiers, UUIDs provide a higher level of uniqueness, generated through standardised algorithms



In the case of the HR dashboard, it is crucial that we use a single, unique, and standardised identifier for every employee!



An analytics dashboard

Data wrangling and cleansing

An introduction

Data wrangling and cleansing are essential processes in data engineering, where raw data is transformed into a structured and usable format for analysis and decision-making.

In the context of our case study, a HR dataset for Credit Bank Corporation could contain data quality issues such as

- Missing values
- Incorrect data types
- Duplicate entries



Case study: Credit Bank Corporation

Data wrangling and cleansing

For data engineers

Using a data wrangling tool like Pandas in Python, a data engineer could:

- Remove duplicate records
- Handle missing values (imputation or deletion)
- Convert data types (e.g. converting strings to dates)
- Standardise formatting (e.g ensuring consistent date formats)
- Identify and remove outliers or anomalies



A data engineer at work

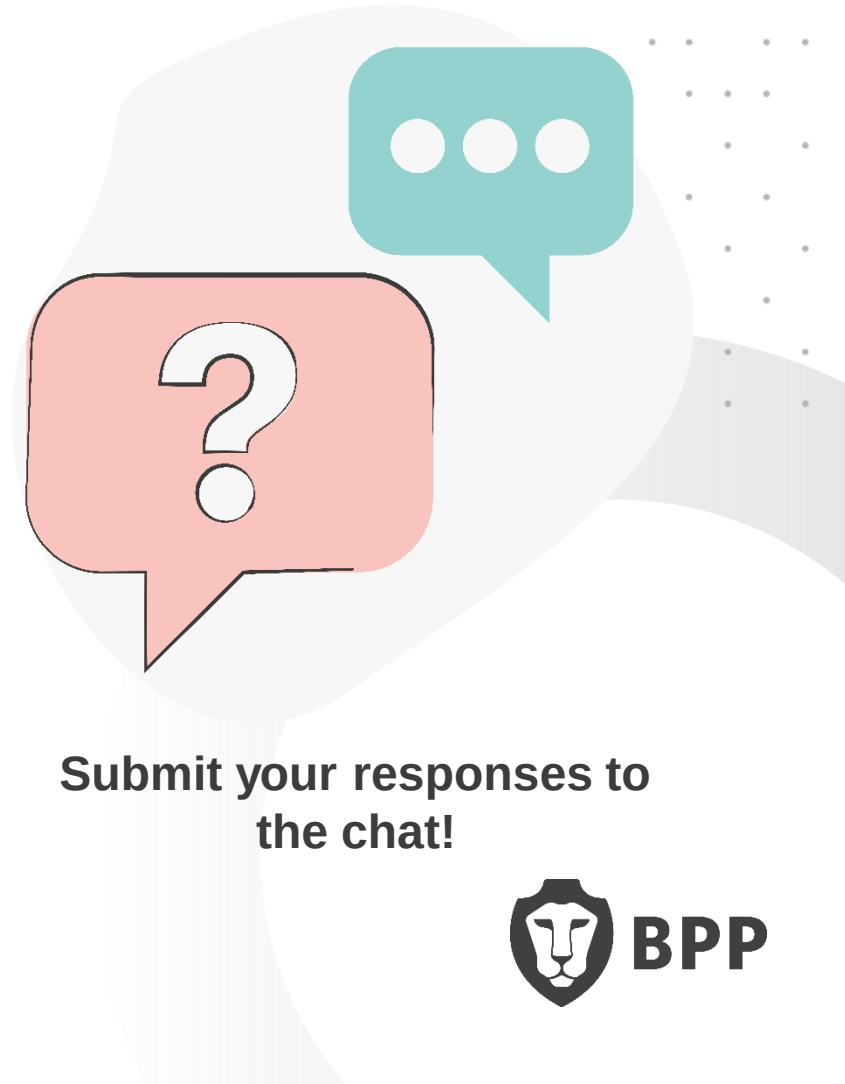
Knowledge Check Poll

Which of the following is NOT a benefit of understanding data lineage?

- A) Ensures transparency in data usage and movement
- B) Aids in understanding patterns of data usage
- C) Enhances trust in data for strategic decision-making
- D) Eliminates the need for data quality checks

Feedback: D – While data lineage provides valuable insights into the lifecycle of data, it does not eliminate the need for data quality checks.

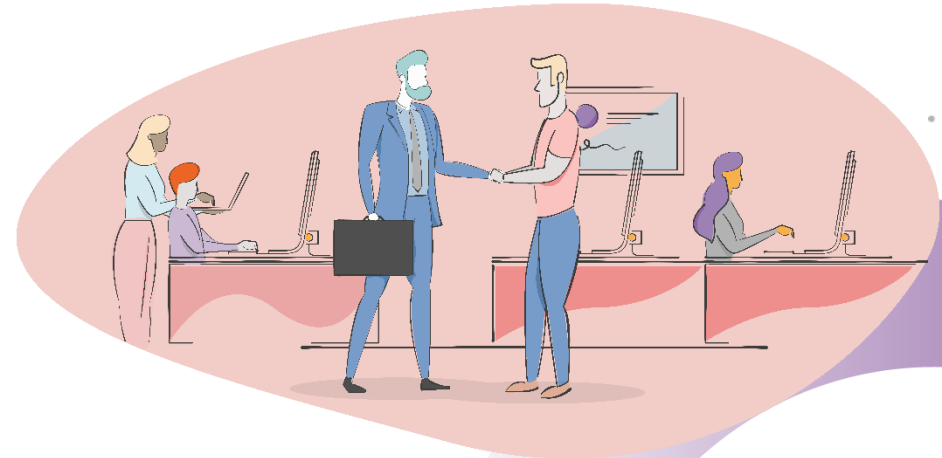
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Focus on Data Stewardship



Section Introduction

Data stewardship

- Data stewardship: Assigns roles for ensuring data accuracy, completeness, reliability
- Importance: Crucial for data quality management
- Best practices: Implementing effective stewardship within organisation

What would data stewardship mean in the context of our case study?



Data stewards are the guardians of **accuracy**, **integrity**, **consistency** and **timeliness** metrics.



Credit Bank Corporation's analytics dashboard

Applying data stewardship

In the context of our case study

The HR team's analytics dashboard vision faces data quality issues, jeopardising key metric integrity and utility.

To combat these issues the data engineering team:

- Explores data quality best practices with data stewardship as the main focus
- Considers a 'hive-mind' approach for refining data quality
- Implements strategies based on standards to ensure more accurate datasets

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Credit Bank Corporation's data engineering team

Data stewardship

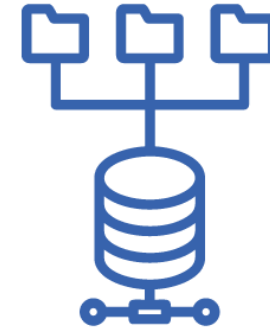
How to implement?



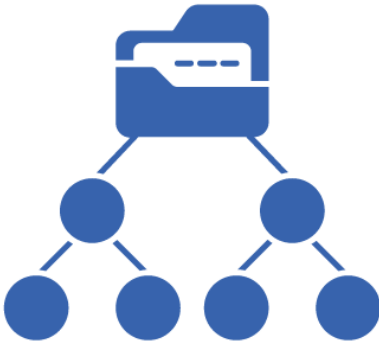
Assignment of Data
Ownership



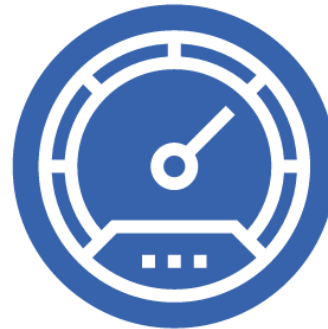
**Data
Stewardship**



Ensuring Data
Provenance



Automated Indexing
of Data



Quality Metrics
Tracking



Regular Data
Audits

Overcoming data challenges with data stewardship

In the context of our case study

Data stewardship techniques that help to overcome challenges in data quality for our HR analytics dashboard, include:



Data cleansing initiatives



Data validation processes



Improvement of data entry procedures



Enhanced data reporting



Feedback loops for continuous improvement



Credit Bank Corporation's HR analytics dashboard

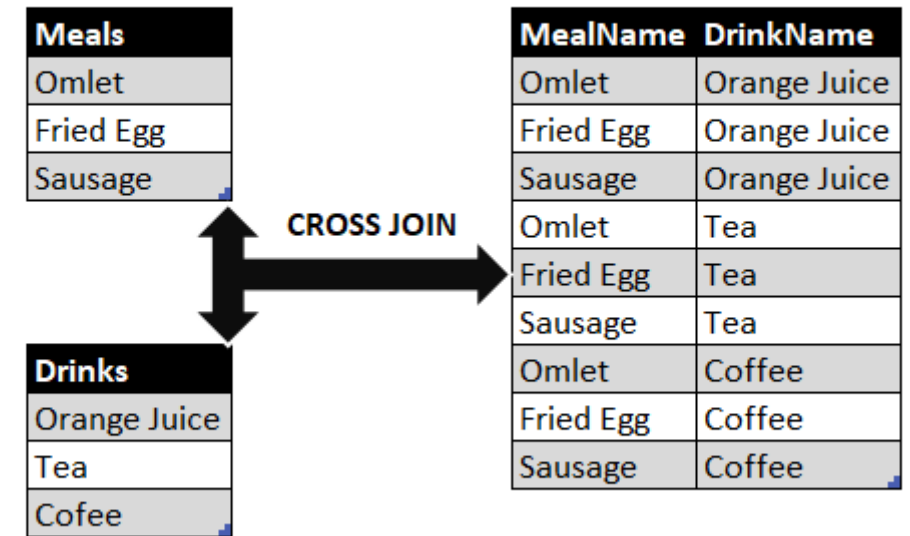
Joining (combining) data

An introduction

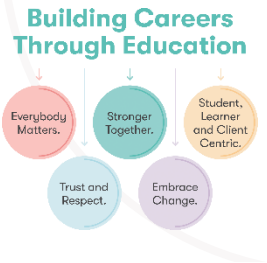
In data engineering, it is often necessary to combine data from multiple sources to provide a unified view, enhancing analysis and decision-making capabilities.

Techniques for joining data may include:

1. Database joins (e.g., inner join, outer join, cross join)
2. Merge operations in data wrangling tools like Pandas
3. Data integration tools like Apache NiFi or Talend



Database join: Cross join



Knowledge check poll

Which of the following is NOT a best practice for data stewardship?

- A) Assignment of data ownership
- B) Ensuring data provenance
- C) Automated indexing of data
- D) Ignoring feedback loops for continuous improvement

Feedback: D – Ignoring feedback loops for continuous improvement is not a best practice for data stewardship.

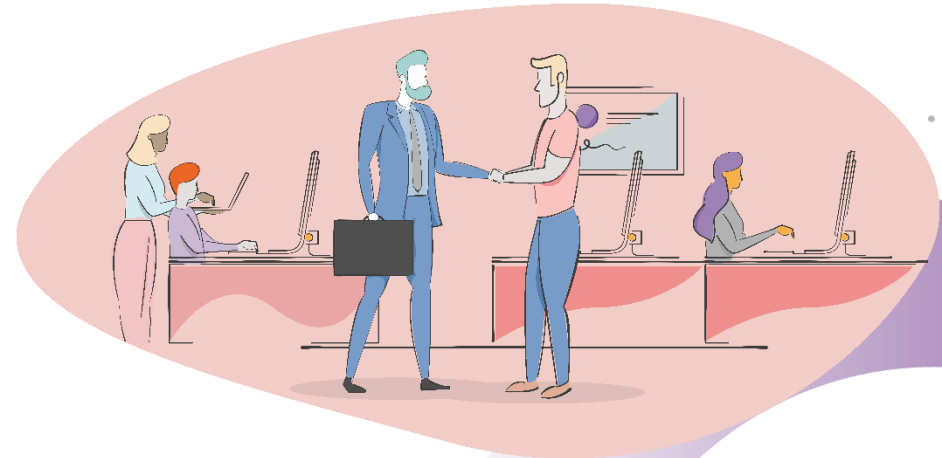
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Focus on Sustainability



Focus on sustainability

Data compression techniques

Here's what you need to know:

- Data compression techniques reduce the environmental impact of data storage and transfer
- Advanced compression algorithms decrease the size of data at rest and in transit
- This reduction leads to lower storage and bandwidth requirements
- Consequently, it results in lower energy consumption and carbon footprint



Brotli Compression: A lossless data compression algorithm developed by Google

Focus on sustainability

Green data transfer

Organisations can also adopt green data transfer practices to minimise their environmental impact.

These practices include:

- Smart Data Transfer Scheduling
- Green Protocols for Data Transfer
- Regular Audits of Data Efficiency



Green Data Transfer

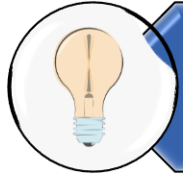
Image source:

InformationSecurityManagement.co.uk

Focus on sustainability

Strategies to support net-zero goals

If you remember to implement the following data quality strategies, you are well on your way to successful data stewardship:



Avoiding data duplication



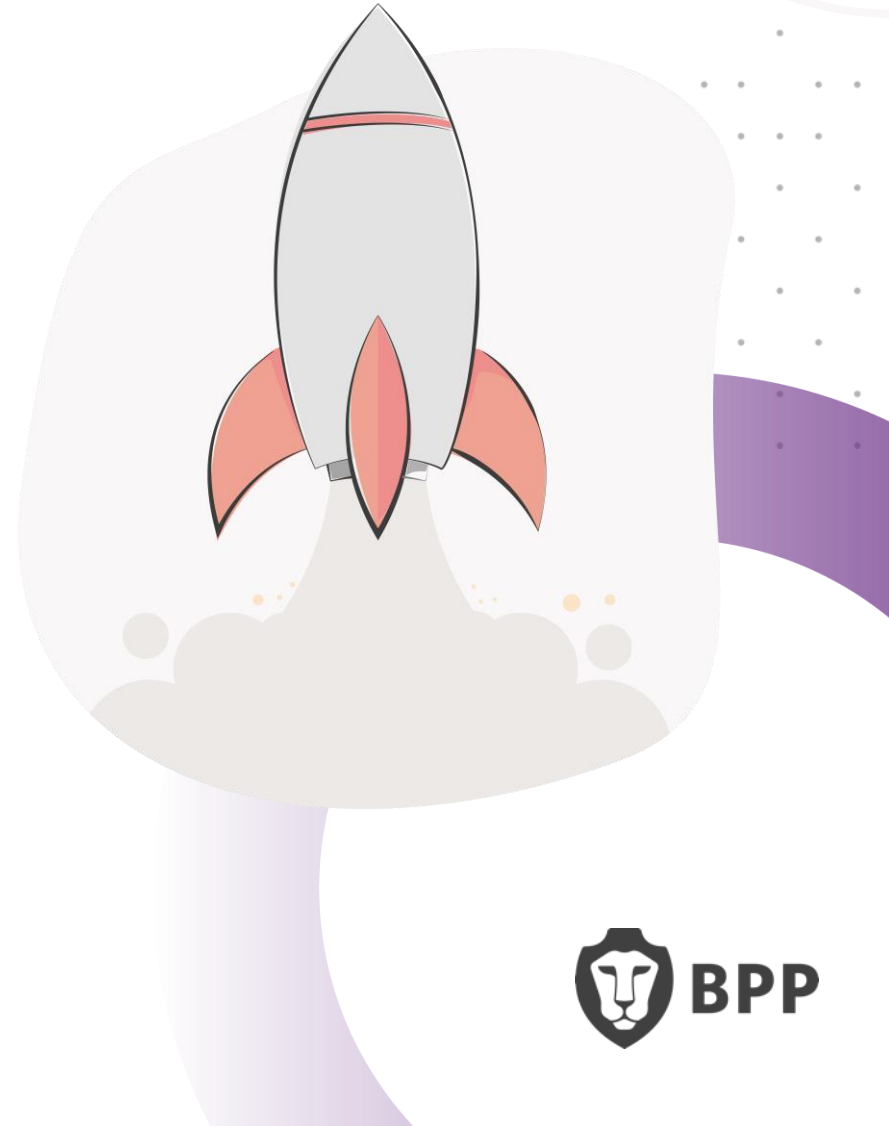
Lean data management



Rationalising data storage



Data lifecycle management



Knowledge check poll

Which of the following is NOT a recommended practice for green data transfer?

- A) Utilising data compression techniques
- B) Scheduling data transfers during peak hours
- C) Choosing energy-efficient transfer protocols
- D) Conducting regular audits of data efficiency

Feedback: B - Scheduling data transfers during peak hours is not a recommended practice for green data transfer.

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Data Quality Management Beyond Data Stewardship



Beyond data quality management

Semantic validity in linked data

In the context of linked data, semantic validity refers to ensuring that data accurately represents the real-world constructs it is intended to model.

This is particularly important for facilitating interoperability and meaningful data integration across diverse datasets.

Semantic validity models include:

- RDF (Resource Description Framework)
- SPARQL
- OWL (Web Ontology Language)



Semantic validity models



Beyond data quality management

Best practices for semantic validation

Here's what you need to know:

- Consistent use of vocabularies and ontologies ensures adherence to predefined terms and definitions
- Enhances interoperability and semantic clarity of data
- Enables effective communication and interpretation across different systems and stakeholders



Semantic validity models

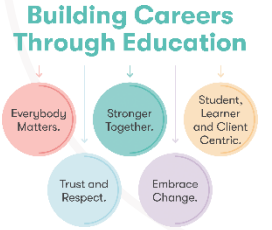
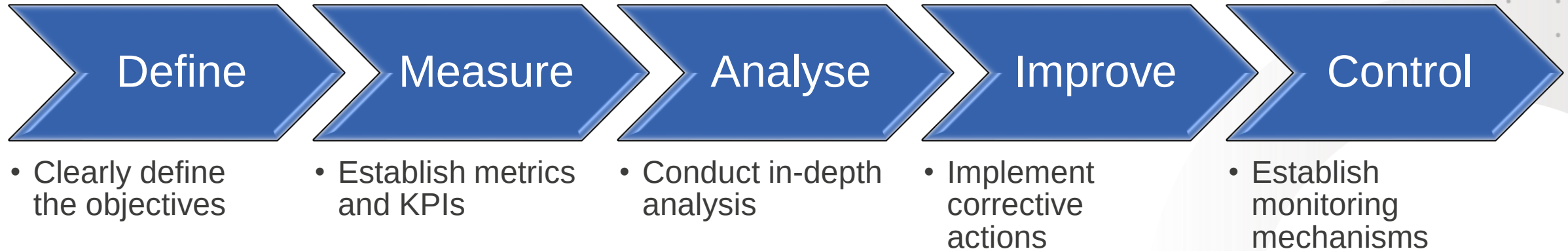


Implementing a data quality framework

A step-by-step guide (DMAIC)

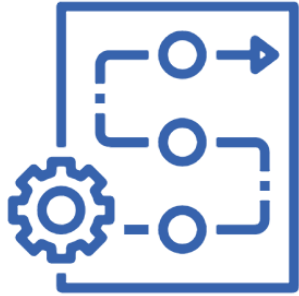
Developing a robust data quality framework is essential for organisations to ensure the reliability, accuracy, and integrity of their data assets.

Here's how:



The role of data governance

Supporting data quality



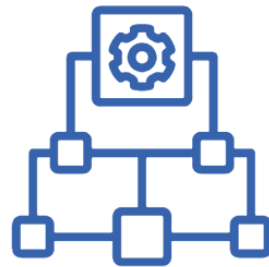
Establishing policies,
procedures, and
standards



**The Role of Data
Governance in Data Quality**



Ensures
accountability and
responsibility



Provides a
framework for
decision-making



Facilitates
collaboration and
communication

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Practical: Data Quality Testing



Apply hands-on exercise

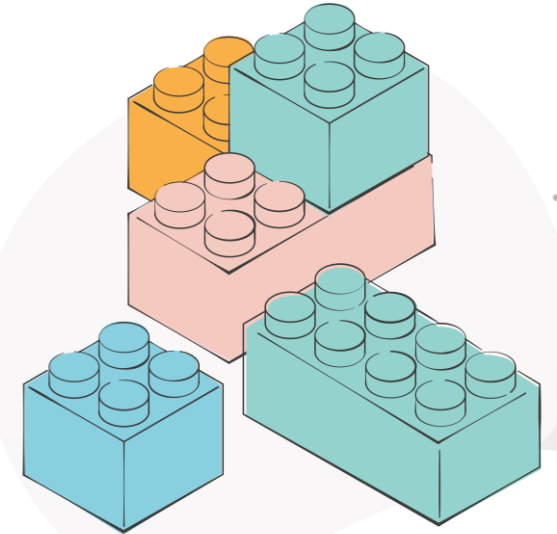
Your homework (part 1)

Your brief is as follows:

1. Find & download IBM HR Analytics dataset from Kaggle
2. Analyse metadata & identify potential data quality issues
3. Prepare data: retain specific columns, create metadata sheets
4. Document findings, upload file to GitHub
5. Review KSBs covered, reflect in learning journal

Step-by-step guidance is provided here: [M1T2 Apply: Intro to data](#)

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Apply exercise

Apply hands-on exercise

Your homework (part 2)

In this exercise, you create a high-quality dataset by consolidating data from three separate low-quality datasets.

These are currently provided as three separate sheets:

- [Mwanza_r, Pwani_r, DSM\(ilala\)_r](#)

You have received the brief below in an email:

```
From: <datascientist@company.com>  
To: <dataengineer@company.com>  
Re: Dataset usability - healthcare facility data
```

```
Good morning, as part of our new healthcare facility project for Tanzania, we would like to perform clustering on the data for the three localities Mwanza, Pwani and Ilala. The data has been sourced from local authorities, they all use slightly different formats. Can you please pool this into a single dataset and make sure any quality issues are addressed. Please also create a schema for the data.
```

```
Many thanks,
```

Step-by-step guidance is provided here: [M1T2 Apply: Intro to data](#)

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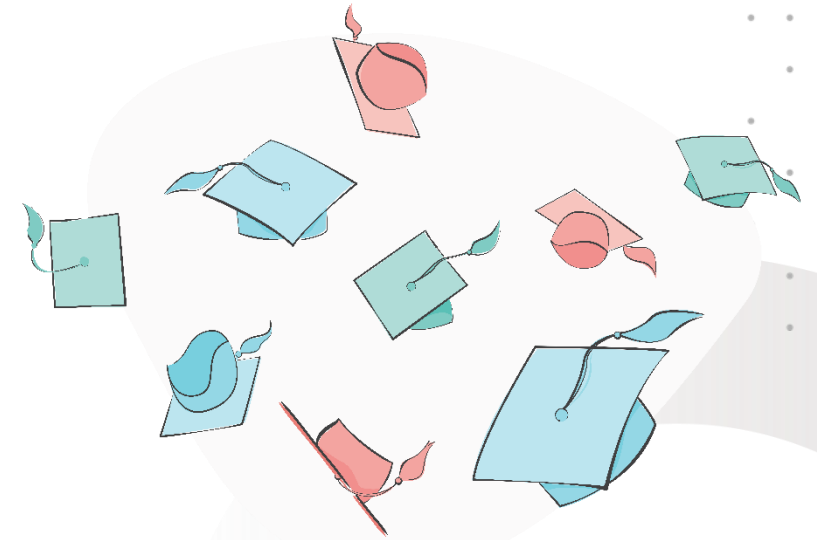


Key Learning Summary

The key takeaways from this session are as follows:

- Data quality encompasses dimensions like accuracy, integrity, consistency, timeliness, and reliability
- Following data and metadata standards like FAIR principles, Dublin Core, etc. helps improve data findability, accessibility, interoperability, and reusability
- Data lineage tracking and use of unique identifiers enable transparent data management and enhance data integrity/compliance
- Data wrangling, cleansing, and stewardship techniques help improve data quality
- Data compression and green data transfer practices aid sustainability and net-zero goals by reducing storage/bandwidth requirements
- Implementing a robust data quality framework (e.g. DMAIC) supported by data governance is key for maintaining high data quality standards

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Thank you

**Do you have any questions,
comments, or feedback?**

